RCS03 | Student Packet

CTS Cohort 6 | CTS Day 11 | 2026-07-22 | 10:00 AM-3:00 PM | 6 students

Electrical Fundamentals: Receptacles and Box Layout

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Big idea: Residential wiring is a controlled path; professionals de-energize, verify, route, terminate, inspect, and document.

Learning targets

- Trace hot, neutral, and equipment-grounding paths in a training circuit
- Lay out two receptacle boxes on the capstone wall
- Prepare and terminate conductors on training receptacles
- Complete visual, mechanical, and continuity checks while de-energized

Today's run

10:00-10:15 | Arrival, PPE, retrieval warm-up

10:15-10:30 | Morning quiz

10:30-10:50 | Quiz review and energy-boundary check

10:50-11:20 | Hot, neutral, and grounding path model

11:20-12:00 | Box-layout and receptacle termination demonstration

12:00-12:30 | Verification ladder guided practice and lab release

12:30-1:00 | Lunch and controlled reset

1:00-2:25 | Two-box, two-receptacle de-energized capstone lab

2:25-2:45 | Visual, tug, continuity, evidence, cleanup

2:45-3:00 | Exit ticket and Day 4 preview

Key words

hot | neutral | equipment grounding conductor | receptacle | device box | sheathing | insulation | terminal | pigtail | continuity

Guided worksheet

1. Trace the training circuit path with arrows.

_____

_____

2. Label hot, neutral, and grounding conductors.

_____

_____

3. Record both box layout measurements.

_____

_____

4. List four termination defects to inspect.

_____

_____

5. Record continuity results only after the instructor confirms de-energization.

_____

_____

Afternoon lab mission

Students mount two training device boxes in their capstone walls, route a de-energized training cable path, prepare conductors, terminate two receptacles, and complete a four-part verification. No connection to building power is permitted.

Safety boundary

- Training mock-up only. Keep every circuit de-energized, isolated from premises wiring, and under instructor control. This class does not authorize unlicensed field electrical work.
- Treat any unverified conductor as energized
- Eye protection during drilling and cable preparation
- No damaged insulation or exposed copper outside terminals
- Continuity mode only on isolated circuits
- Instructor controls any test source

Build / test sequence

[ ] Energy-control briefing

[ ] Verify training source isolated

[ ] Lay out two boxes

[ ] Instructor layout approval

[ ] Mount boxes

[ ] Route training cable

[ ] Prepare conductors

[ ] Terminate receptacles

[ ] Visual and tug inspection

[ ] De-energized continuity check

[ ] Label, photograph, reset

Evidence record

[ ] Circuit-path worksheet: _____

[ ] Box-layout measurements: _____

[ ] Two completed receptacle terminations: _____

[ ] Visual/tug/continuity log: _____

[ ] Instructor approval record: _____

Reflection and exit ticket

Draw the training circuit path

_____

_____

Name one strong termination feature

_____

_____

State the meter safety boundary

_____

_____

Name one inspection you completed

_____

_____

Instructor verification

Status: [ ] Approved [ ] Correct and recheck [ ] Needs follow-up

Instructor initials: _____ Date/time: _____

RCS03 | Morning Quiz

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Name: _____ Date: _____

Answer from your own understanding. This is a teaching check, not a trap.

1. What conductor normally carries current from the source to the load?

_____

_____

2. What conductor normally provides the return path?

_____

_____

3. What is the purpose of the equipment-grounding conductor?

_____

_____

4. What must be proven before continuity mode is used?

_____

_____

5. What is a tug test checking?

_____

_____

6. Why must copper not extend loosely outside a terminal?

_____

_____

7. May this mock-up be connected to building power?

_____

_____

8. Name the four verification stages used today.

_____

_____

9. Who approves the box layout before drilling?

_____

_____

10. Does course practice authorize unlicensed field electrical work?

_____

_____